Eur. J. Biochem. **266**, 820–830 (1999) © FEBS 1999

Azide binding to the trinuclear copper center in laccase and ascorbate oxidase

I. Gromov[1,*], A. Marchesini[2], O. Farver[3], I. Pecht[4] and D. Goldfarb[1]

[1]*Department of Chemical Physics, Weizmann Institute of Science, Rehovot, Israel;* [2]*Instituto Sperimentale per la Nutrizione Della Piante, Torino, Italy;* [3]*Department of Analytical and Pharmaceutical Chemistry, Royal Danish School of Pharmacy, Copenhagen, Denmark;* [4]*Department of Immunology, Weizmann Institute of Science, Rehovot, Israel*

Azide binding to the blue copper oxidases laccase and ascorbate oxidase (AO) was investigated by electron paramagnetic resonance (EPR) and pulsed electron-nuclear double resonance (ENDOR) spectroscopies. As the laccase : azide molar ratio decreases from 1 : 1 to 1 : 7, the intensity of the type 2 (T2) Cu(II) EPR signal decreases and a signal at $g \approx 1.9$ appears. Temperature and microwave power dependent EPR measurements showed that this signal has a relatively short relaxation time and is therefore observed only below 40 K. A $g \approx 1.97$ signal, with similar saturation characteristics was found in the AO : azide (1 : 7) sample. The $g < 2$ signals in both proteins are assigned to an $S = 1$ dipolar coupled Cu(II) pair whereby the azide binding disrupts the anti-ferromagnetic coupling of the type 3 (T3) Cu(II) pair. Analysis of the position of the $g < 2$ signals suggests that the distance between the dipolar coupled Cu(II) pair is shorter in laccase than in AO. The proximity of T2 Cu(II) to the $S = 1$ Cu(II) pair enhances its relaxation rate, reducing its signal intensity relative to that of native protein. The disruption of the T3 anti-ferromagnetic coupling occurs only in part of the protein molecules, and in the remaining part a different azide binding mode is observed. The 130 K EPR spectra of AO and laccase with azide (1 : 7) exhibit, in addition to an unperturbed T2 Cu(II) signal, new features in the $g_{\|}$ region that are attributed to a perturbed T2 in protein molecules where the anti-ferromagnetic coupling of T3 has not been disrupted. While these features are also apparent in the AO : azide sample at 10 K, they are absent in the EPR spectra of the laccase : azide sample measured in the range of 6–90 K. Moreover, pulsed ENDOR measurements carried out at 4.2 K on the latter exhibited only a reduction in the intensity of the 20 MHz peak of the ^{14}N histidine coordinated to the T2 Cu(II) but did not resolve any significant changes that could indicate azide binding to this ion. The lack of T2 Cu(II) signal perturbation below 90 K in laccase may be due to temperature dependence of the coupling within the trinuclear : azide complex.

Keywords: blue copper oxidases; EPR; ENDOR; trinuclear copper center; anti-ferromagnetic coupling.

Ascorbate oxidase (AO) and laccase are blue copper oxidases that catalyze oxidation of a substrate with the concomitant four electron reduction of oxygen [1,2]. They contain four copper binding sites per functional unit; a type 1 (T1) site which is involved in electron uptake from the substrate, a type 2 (T2) site, and an electron paramagnetic resonance (EPR) silent type 3 (T3) site which, in the oxidized state, consists of a pair of anti-ferromagnetically coupled Cu(II) ions [1,2]. The latter three Cu(II) binding sites are proximal and constitute a trinuclear center which is the catalytic dioxygen reduction site [3,4]. Although the 3D structures of AO and its derivatives have considerably advanced our understanding of its reaction mechanism [5], the details of the dioxygen reduction at the trinuclear center remain, however, only partly comprehended.

Correspondence to D. Goldfarb, Department of Chemical Physics, Weizmann Institute of Science, Rehovot, Israel. Fax: + 972 8 344123, E-mail: cigoldfa@wis.weizmann.ac.il

Abbreviations: EPR, electron paramagnetic resonance; ENDOR, electron-nuclear double resonance; lac, laccase; AO, ascorbate oxidase; ZFS, zero field splitting; T1, type 1; T2, type 2; T3, type 3; MCD, magnetic circular dichroism; CW, continuous wave; ED, echo-detected; HS, hyperfine-selective; ESEEM, electron-spin echo envelope modulation; G, gauss; MW, microwave; RF, radio-frequency.

(Received 22 June 1999, accepted 28 September 1999)

One of the approaches employed for the examination of oxygen interaction with the trinuclear center has been the investigation of exogenous ligand binding such as peroxide, azide, fluoride and cyanide [5–13].

The 3D structure of oxidized AO shows that the average distance between the copper ions in the trinuclear center is 0.374 nm with an oxygen ligand bridging the two T3 coppers, defined as Cu2 and Cu3, as shown in Fig. 1 [3,4]. The 3D structure of the azide derivative of oxidized AO shows that two azide ions coordinate terminally to one of the T3 copper ions (Cu2), causing the removal of the bridging oxygen ligand and increasing the Cu2–Cu3 distance to 0.51 nm [5]. The distance from the putative T2 (Cu4) increases to 0.46 nm for Cu2–Cu4 while that of Cu3–Cu4 remains short (0.36 nm) [5], thus suggesting that the EPR active Cu(II) may no longer be Cu4. The 3D structure of T2 depleted AO shows some occupancy in the Cu4 site along with partial depletion of Cu2 and Cu3. The EPR spectrum, however, reveals no indication for the presence of a T2 signal in these crystals [14]. This suggests that the three Cu(II) ions in the trinuclear center are exchangeable and each pair may become anti-ferromagnetically coupled, possibly accounting for the heterogeneity proposed earlier for T2 depleted laccase [15]

Comparison of the amino acid sequences of laccase from various sources and AO shows that their structures are closely

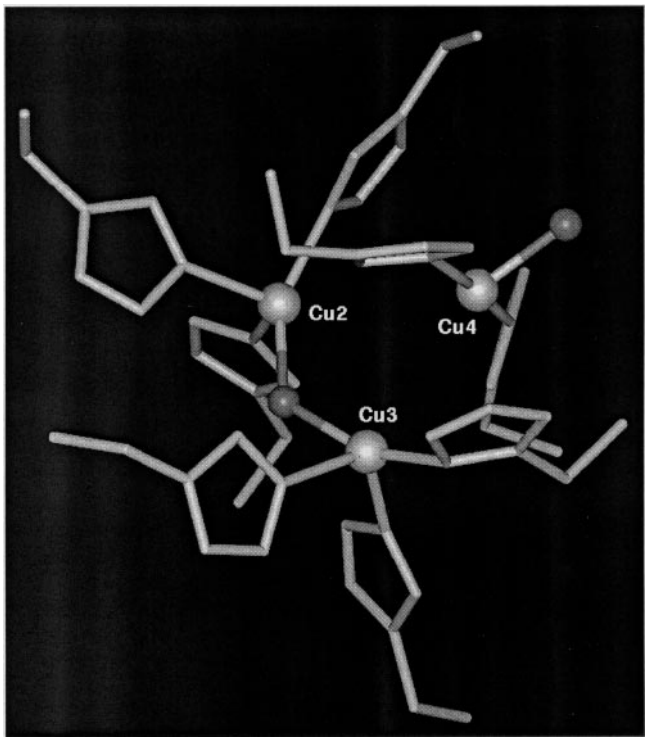


Fig. 1. The structure of the trinuclear copper center of ascorbate oxidase. Data from [4].

related [16]. This was recently confirmed by the reported 3D structure of T2 depleted laccase from the fungus *Coprinus cinerus* [17]. In contrast to T2 depleted AO, a full occupancy of copper ions in T3 was observed. The presence of the trinuclear center in laccase was first proposed on the basis of functional [6,18–20], and EPR studies [8]. More recently, magnetic circular dichroism (MCD) measurements of azide binding [9,10], led to a model, where in laccase, the first azide ion binds to the trinuclear center by bridging T2 and one of the T3 Cu(II) ions, whereas the second azide ligand coordinates with a lower affinity, bridging T2 and the other T3 copper ion [11]. In addition, it was reported that in less than 10% of the azide bound sites, the endogenous oxo-bridge between the anti-ferromagnetically coupled T3 Cu(II) is disrupted leading to an $S = 1$ species with $g = 1.86$ [11]. Similar investigations were also carried out on AO and three coexisting modes of azide binding were proposed [12]. In the first the azide bridges T2 and T3, and in the other two it binds terminally to either T2 or T3. A $g < 2$ signal, indicative of T3 bridge disruption upon azide binding, was not observed for AO. The Cu–N_3^- vibrations in laccase have been recently measured by Fourier transform IR and resonance Raman spectroscopy [21] but the frequency observed did not allow resolving between a model where the azide bridges the T2 and T3 sites or is terminally bound.

The inconsistencies between the modes of azide binding to AO emerging from X-ray crystallography and the interpretation of the EPR/MCD results of laccase and AO prompted us to carry out microwave power and temperature dependent EPR, and electron-nuclear-double-resonance (ENDOR) measurements of the azide complexes of laccase and AO. The EPR measurements were designed to distinguish between the various Cu(II) sites according to their relaxation properties and from the ENDOR experiments we expected to resolve explicitly Cu(II)-N_3^- hyperfine interactions [13]. Two major binding modes were observed. In the first, the azide binding causes the disruption of the anti-ferromagnetic coupling of T3 and generates a dipolar coupled $S = 1$ Cu(II) pair in both proteins. In the second mode, the coupling of T3 is not disrupted and the

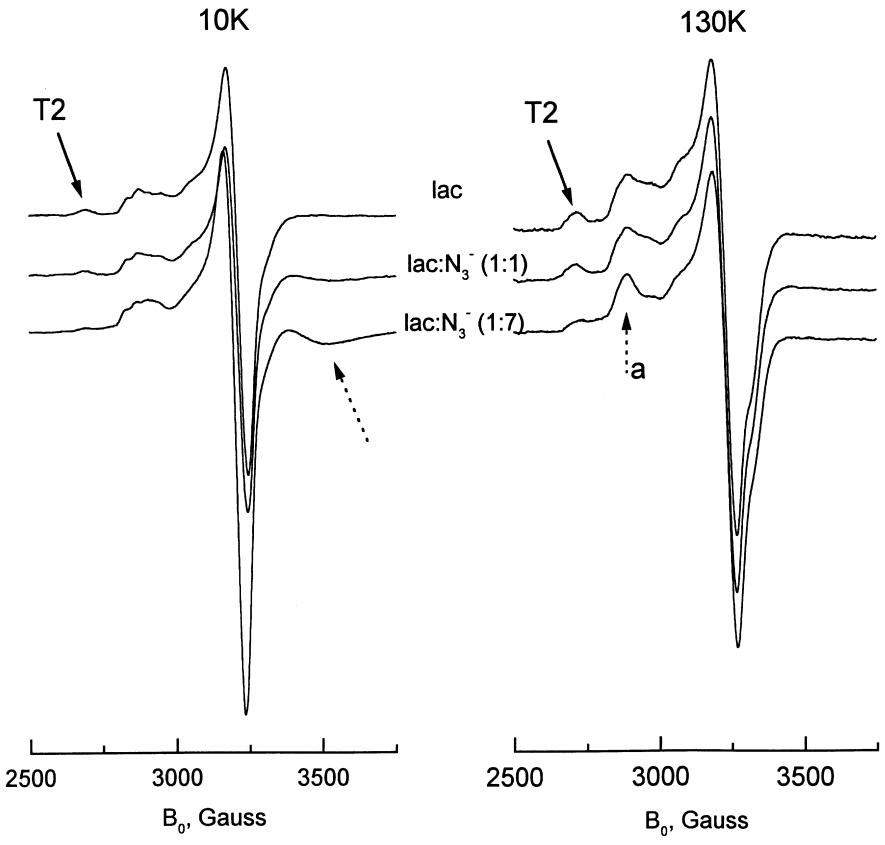


Fig. 2. CW-EPR spectra of laccase, and laccase : azide (1 : 1 and 1 : 7) solutions (1.5 mM, pH = 6) recorded at 130 K (ν = 9.178 GHz) and 10 K (ν = 9.125 GHz). Both spectra where recorded with $P = -10$ dB where 0 dB = 258 mW. The solid arrows mark the position where only T2 contributes to the spectrum and the dotted arrows mark the signal of the new species. a, shows significant increased intensity around 2890 G.

observed new features in the EPR spectra suggest the formation of perturbed T2 caused by azide binding. Surprisingly, these features do not appear in laccase : azide EPR spectra recorded below 130 K. In addition, 4.2 K ENDOR measurements carried out on laccase with and without azide did not show any significant change in the close environment of T2 that could be attributed to azide binding. This unusual behavior is explained in terms of a temperature dependent magnetic coupling among the three Cu(II) ions coppers in the trinuclear : azide complex.

MATERIALS AND METHODS

Sample preparation

Laccase. Laccase (Japanese lacquer tree, *Rhus vernicifera*) was purified according to published procedures [20,22], and two samples with 1 : 1 and 1 : 7 molar ratio of protein : azide were prepared. The final concentration of the protein in both samples was 1.5 mM (potassium phosphate buffer, pH 6.0). The NaN_3 solutions (0.023 M and 0.15 M, respectively) were added at room temperature, incubated at 4 °C for 24 h and stored in liquid nitrogen until the measurements were performed. The ratio of 1 : 7 was chosen to ensure that maximum binding was achieved [11].

Ascorbate oxidase. AO was purified from green zucchini according to reported procedures [7] and a 1 : 7 sample of AO : azide was prepared as described above. The final protein concentration was 1.3 mM (potassium phosphate buffer pH = 6.0). Upon azide addition both the AO and laccase samples turned greenish.

Spectroscopic measurements. Continuous wave (CW) EPR measurements were performed on a Varian E-12 EPR spectrometer at 9.0–9.3 GHz. The spectra were recorded in the 6–100 K temperature range using an Air-Products Model LTR Helitran liquid He cooling system. The frequency was measured with a HP5350B microwave frequency counter and the g-factor was determined using DPPH as a reference. The maximum output power of the bridge was 258 mW (0 dB). All EPR spectra were recorded with a modulation amplitude of 10 Gauss (G).

Pulsed ENDOR measurements were performed at 4.2 K using a home built pulsed EPR spectrometer operating at 9–9.5 GHz [23,24], with a probehead based on a bridged

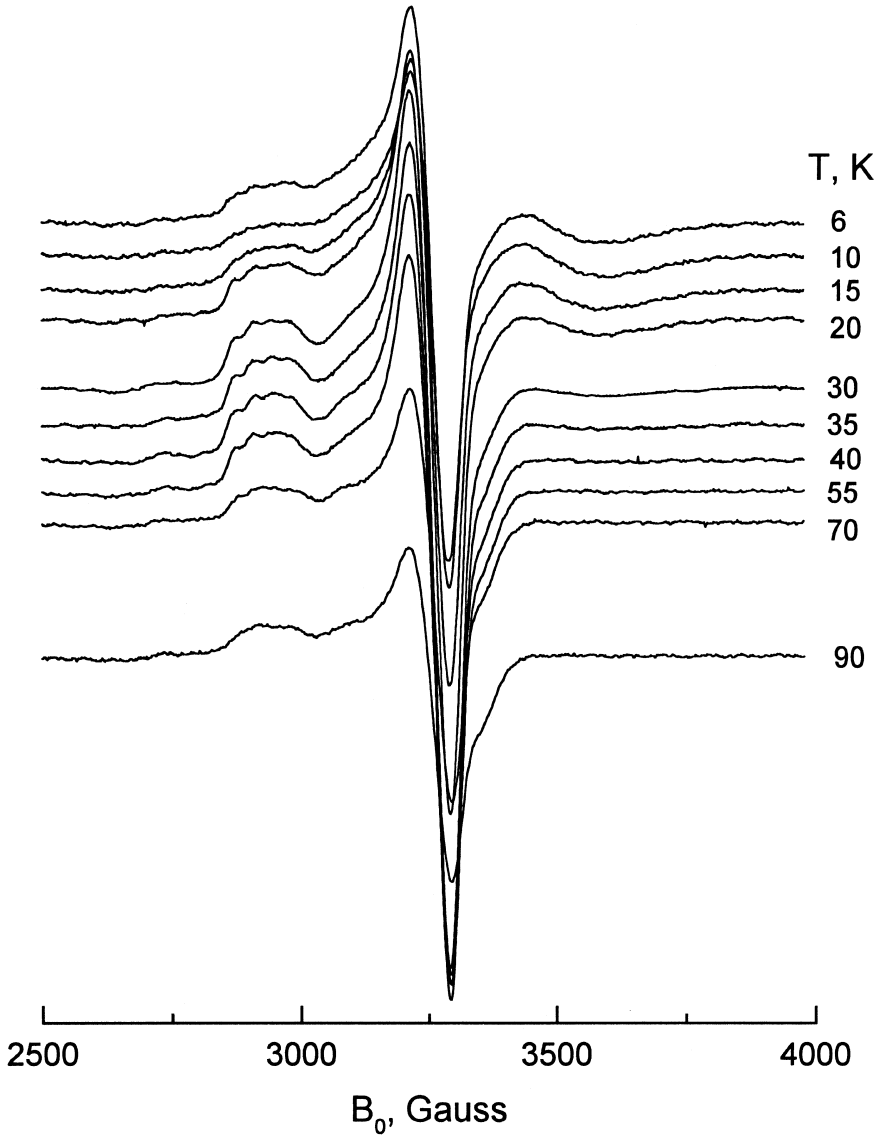


Fig. 3. CW-EPR spectra of a laccase : azide (1 : 7) solution (1.5 mM, pH = 6) recorded at various temperatures (ν = 9.288 GHz, P = 0 dB). The temperature corresponds to that indicated by the temperature controller and a ± 5 K error is expected above 50 K.

loop-gap resonator whose basic features are described elsewhere [24,25]. ENDOR spectra were recorded using the Davies ENDOR sequence [27]:

$$\text{MW } \pi - T - \tfrac{\pi}{2} - \tau - \pi - \tau - echo$$
$$\text{RF } \quad \pi$$

In the hyperfine-selective ENDOR experiment [26] the frequency of the first microwave (MW) π pulse, ν_1 was different from that of the last two echo forming pulses, ν_2, where $\Delta = \nu_1 - \nu_2$. The ENDOR spectrum is acquired by recording the echo amplitude as a function of the frequency of the radio-frequency (RF) pulse. In the HS-ENDOR experiment ENDOR signals appear only when Δ is equal to the hyperfine coupling.

RESULTS

EPR measurements

Laccase. The X-band CW-EPR spectra of laccase recorded at 10 and 130 K are shown in the top traces of Fig. 2, where the characteristic features of the T1 and T2 sites are clear. The feature at the lowest field edge of the $g_{\|}$ region (marked with an arrow and labeled T2) corresponds to the T2 $M_I = -3/2$ component of $A_{\|}$. It does not overlap with the T1 spectrum and is therefore used as an indicator for changes occurring at the T2 site. The spectra recorded at both temperatures show that as the azide concentration increases the T2 signal decreases. Nonetheless, the effect of azide binding is manifested differently in

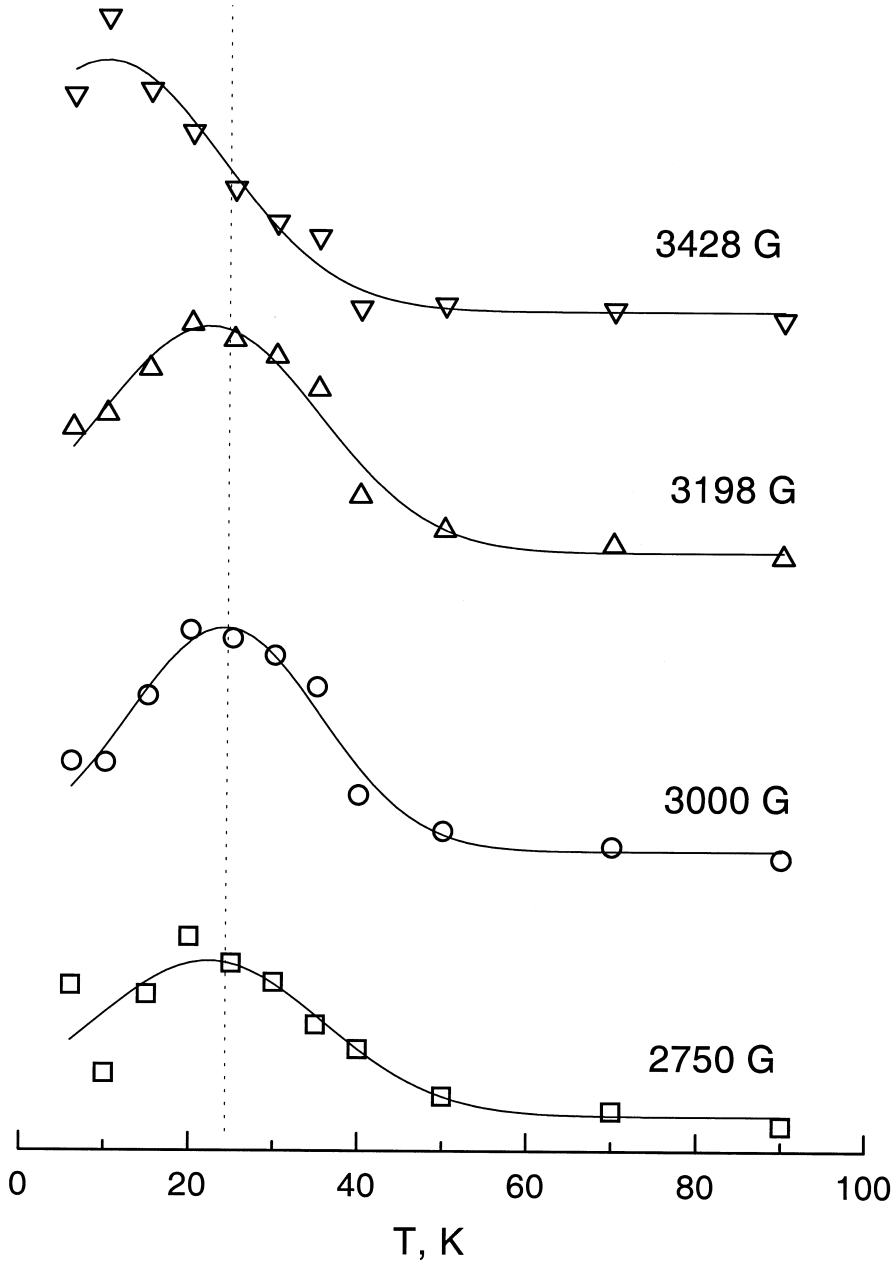


Fig. 4. Plots of the EPR integrated intensity of a laccase : azide (1 : 7) solution (1.5 mM, pH = 6) at different magnetic fields as a function of temperature for $P = -10$ dB. In each curve the maximum intensity was normalized to 1.

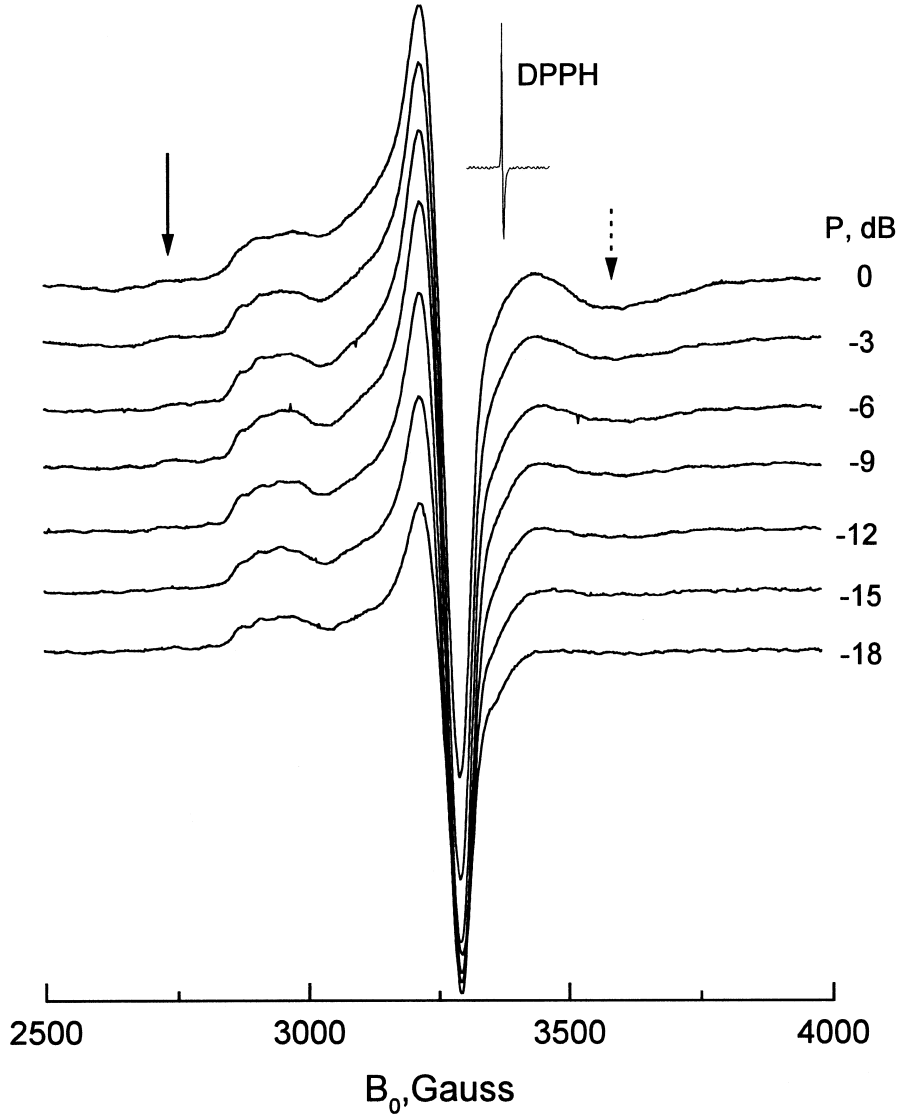


Fig. 5. CW-EPR spectra of a laccase : azide (1 : 7) solution (1.5 mM, pH = 6) recorded at 10 K as a function of microwave power (ν = 9.288 GHz). The solid arrow marks the position where only T2 contributes to the spectrum and the dotted arrow marks the signal of the $S = 1$ species.

spectra recorded at 130 and 10 K. At 130 K the spectrum of the 1 : 7 sample shows a significant increased intensity around 2890 G (see arrow marked with a in Fig. 2). This can be assigned to an $A_{\|}$ feature of a new T2 species which was reported earlier [8,11], and will be hereafter referred to as T2(lac) : N_3^-. This feature, however, does not appear in the 10 K spectrum (Fig. 3). The spectra recorded at 10 K show that the addition of azide also generates a new broad signal at $g \approx 1.9$ (broken arrow in Fig. 2). This signal has already been reported and assigned to a dipolar coupled Cu(II) pair with $S = 1$ arising from the disruption of the oxygen bridge responsible for the T3 anti-ferromagnetic coupling [11]. Unfortunately, we were not able to detect a $\Delta M_S = 2$ signal at half field.

The temperature dependence of the EPR spectrum of the laccase : azide 1 : 7 sample in the range 6–90 K, recorded with maximum available power (258 mW), is shown in Fig. 3. At 20–55 K, all four $A_{\|}$ components of T1 are well resolved, and the T2(lac) : N_3^- feature is absent throughout the 6–90 K range. The $g \approx 1.9$ signal intensity increases with decreasing temperature while the T2 $A_{\|}$ ($M_I = -3/2$) signal has maximum intensity at $\approx$ 35 K. The bulk magnetization of paramagnetic systems is inversely proportional to the temperature as expected from the Curie law [28]. Therefore, under conditions far from saturation, where the saturation factor, $s = [1/(1 + H_1^2\gamma^2 T_1 T_2)] \approx 1$, the EPR signal increases as $1/T$ [28,29]. H_1 is the applied microwave field, and T_1 and T_2 are the spin–lattice and spin–spin relaxation times, respectively. The appearance of a maximum at $\approx$ 35 K shows that at this temperature T_1 becomes sufficiently long so the condition $s \approx 1$ no longer holds and saturation takes place. In contrast, the T_1 of the $g \approx 1.9$ species remains short enough throughout the examined temperature range and saturation is not observed.

Figure 4 shows explicitly the temperature dependence of the EPR signal intensity, measured at a lower power ($P = -10$ dB), at four different magnetic fields, 2750, 3000, 3198 and 3428 G. The intensities were determined from the integrated EPR spectra and the maximum amplitude was normalized to 1. While the signals at the two extreme fields (3428 G and 2750 G) correspond to the $S = 1$ Cu(II) pair and T2, respectively, the signals at the other two fields reflect a superposition of T2 and T1, although at 3000 G the T1 contribution is dominant. Figure 4 shows that while the T1 and T2 signals at the center of the spectrum and the low field T2 signal exhibit a

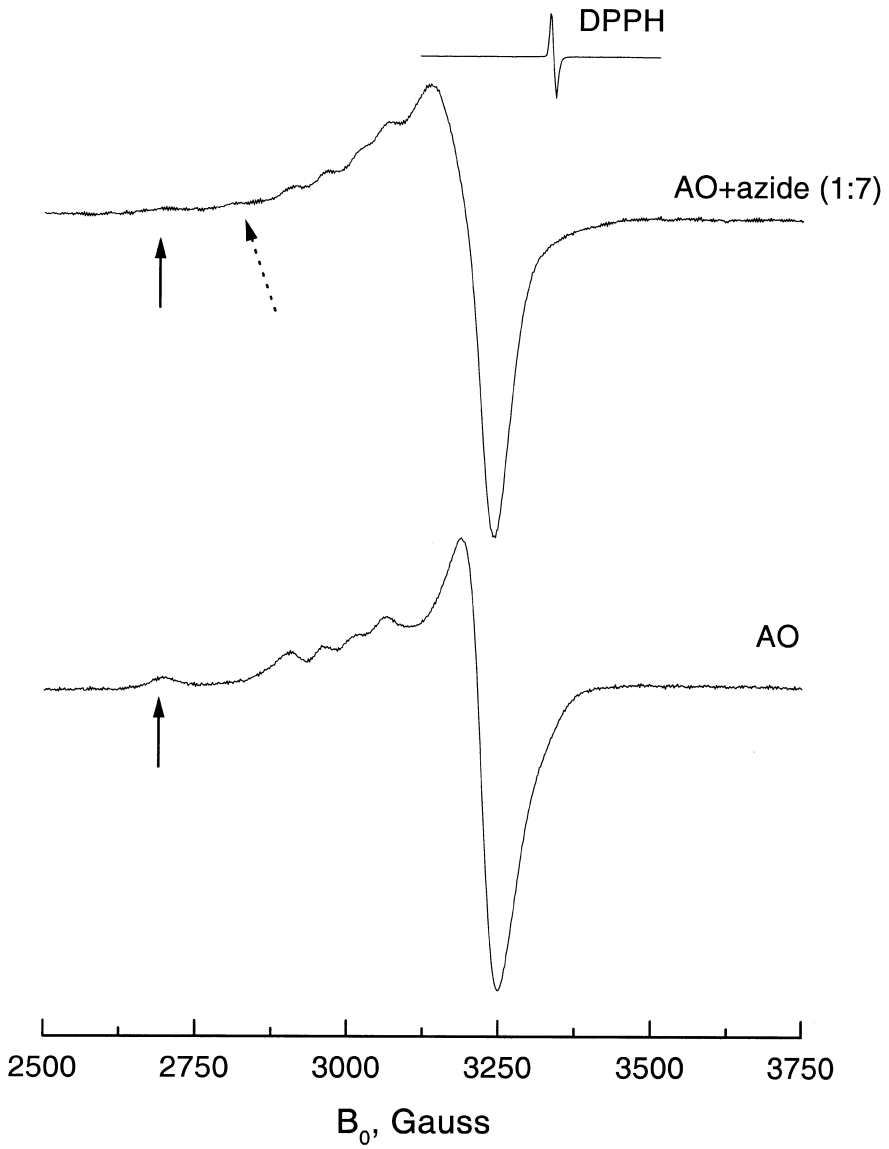


Fig. 6. CW-EPR spectra of AO and AO : azide (1 : 7) solutions (1.3 mM, pH = 6) recorded at 80 K, ν = 9.277 GHz. The solid arrows mark the position where only T2 contributes to the spectrum and the dotted arrow marks the position of the new signals.

maximum at 20–35 K, the $S = 1$ signal does not reach saturation confirming its shorter T_1. The power dependence of the EPR spectrum at 10 K is shown in Fig. 5. Again, the $g \approx 1.9$ signal (denoted by a dashed arrow) keeps increasing with applied power, showing that it is not saturated even at maximum power, whereas the T1 signal begins to saturate between −6 and −9 dB and the T2 signal, at 2750 G (denoted by a solid arrow) at ≈ −3 dB.

Ascorbate oxidase. The spectrum of native AO, recorded at 80 K, is shown in the bottom trace of Fig. 6. The hyperfine structure at the $g_{\|}$ region of T1 is well resolved and the low field signal (2750 G) of T2 is clear (see solid arrow). The addition of azide (1 : 7 molar ratio) leads to a decrease in the low field T2 signal, together with the appearance of a new feature at 2860 G (marked with a dashed arrow in Fig. 6) and a broad signal at 3000–3200 G. The signal at 2860 G is attributed to an $A_{\|}$ resolved singularity of a perturbed T2 referred to as T2(AO) : N_3^- which has already been reported [12]. The 14 K spectra of AO and AO : azide (1 : 7) recorded at three different power levels are presented in Fig. 7. A new signal, the intensity of which increases with increasing power, appears at $g \approx 1.97$ in the spectrum of AO : azide (1 : 7). This signal is not detectable above 40 K even at full microwave power. Based on its low g-value and the temperature and power dependence characteristics, which are similar to those exhibited by the $g \approx 1.9$ signal in laccase : azide (1 : 7), we assign it to be a $S = 1$ Cu(II) pair too. The low $g < 2$ value rules out the assignment of this signal to the perturbed T2. Both 14 K and 80 K spectra show that azide binding is associated with a decrease in the T2 signal and the appearance of a new signal of T2(AO) : N_3^-.

Pulsed ENDOR measurements

ENDOR measurements were carried out on laccase and its azide complexes in an attempt to detect ^{14}N signals of the bound azide or other changes that may occur in the coordination sphere of T2 due to azide binding. Figure 8 shows Davies ENDOR spectra of laccase measured at 4.2 K at various magnetic fields along the EPR powder pattern. The spectra were measured with nonselective microwave pulses to

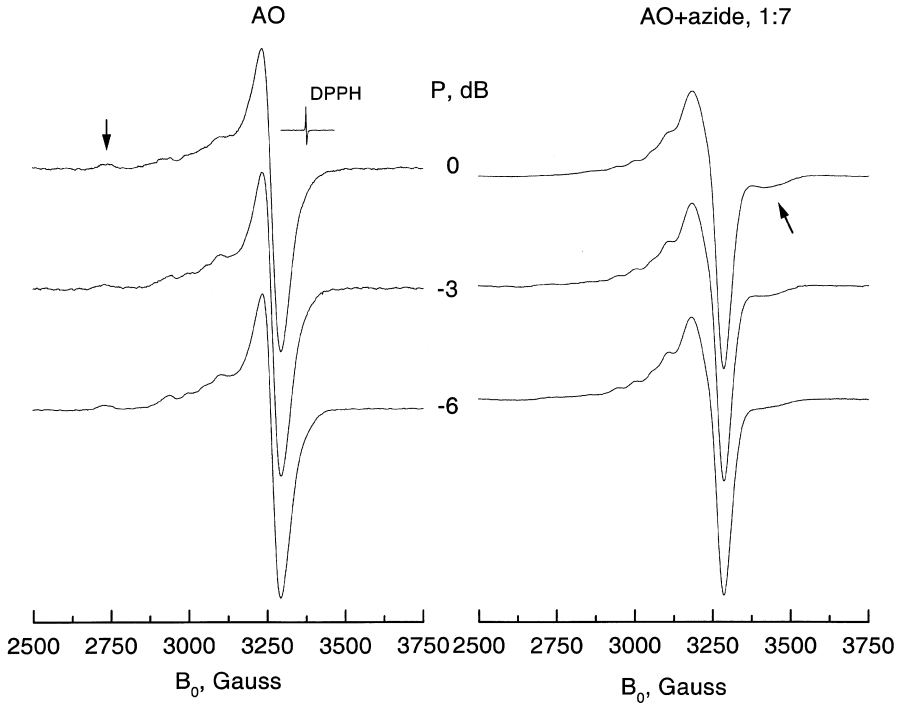


Fig. 7. CW-EPR spectra of AO and AO : azide (1 : 7) solutions (1.3 mM, pH = 6) recorded at 14 K, ν = 9.277 GHz and three different microwave power levels. The arrow in the AO spectrum points to the T2 feature and in the AO-azide sample to the $g < 2$ signal.

reduce signals of weakly coupled protons [30]. At the highest field edge of the echo-detected (ED) EPR spectrum (3320 G) only T2 contributes to the echo and the ENDOR spectrum consists of a broad single line (width of $\approx$ 8 MHz) centered at 20 MHz. Strongly coupled ^{14}N nuclei, the hyperfine coupling of which, A_N, is larger than twice the ^{14}N Larmor frequency, ν_I, exhibit a doublet centered at $A_N/2$ split by $2\nu_I$ ($\approx$ 2 MHz at the magnetic fields employed). In addition, each doublet component should be split by the ^{14}N nuclear quadrupole interaction [30]. None of these splittings are resolved due to the inhomogeneous broadening characteristic of orientationally disordered systems. The 20 MHz line is assigned to histidine nitrogens directly bound to the T2 Cu(II) site with A_N = 40 MHz.

The ENDOR spectrum of laccase measured at 3220 G, which corresponds to maximal echo intensity, near the $g_\perp$ positions of T1 and T2, exhibit overlapping ^{1}H and ^{14}N signals of the T1 and T2 centers. The ^{1}H and ^{14}N signals

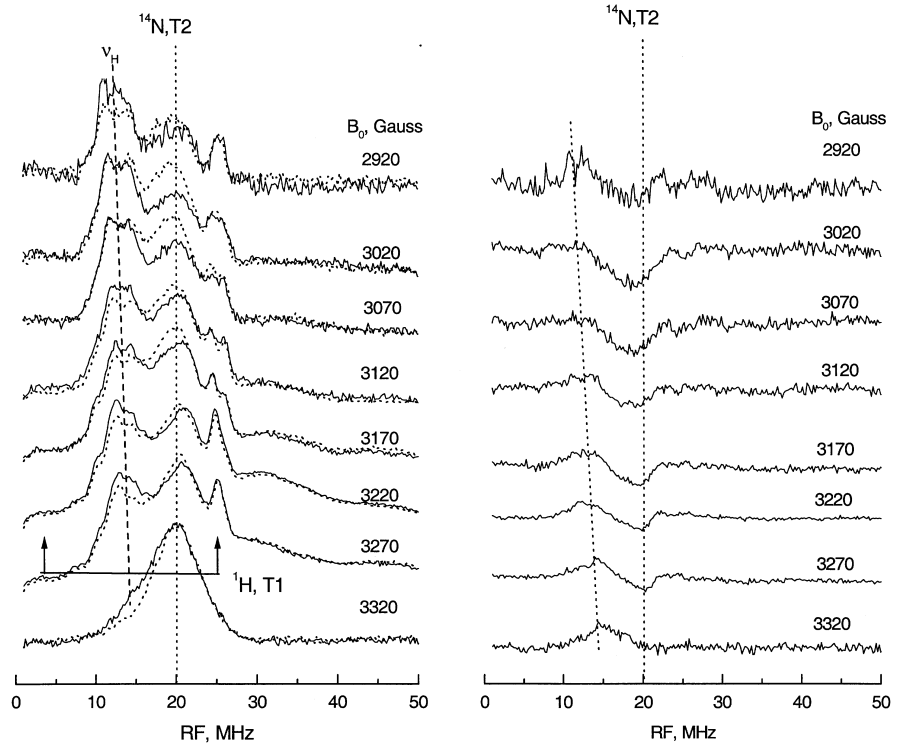


Fig. 8. Davies ENDOR spectra of a laccase solution (1.5 mM, pH = 6) (dotted lines) and laccase : azide (1 : 7) (solid lines) recorded at 4 K at different magnetic fields within the EPR powder pattern (left panel) and the difference spectra are shown on the right hand side of the figure. The experimental conditions: ν = 9.40 GHz, radio-frequency pulse length, 3.5 μs, and microwave pulse lengths 0.04, 0.02, 0.04 μs. The proton's Larmor frequency is indicated by a slanted dotted line and the position of the protons doublet is indicated as well. The dashed line represent the position of the T2 ^{14}N signal.

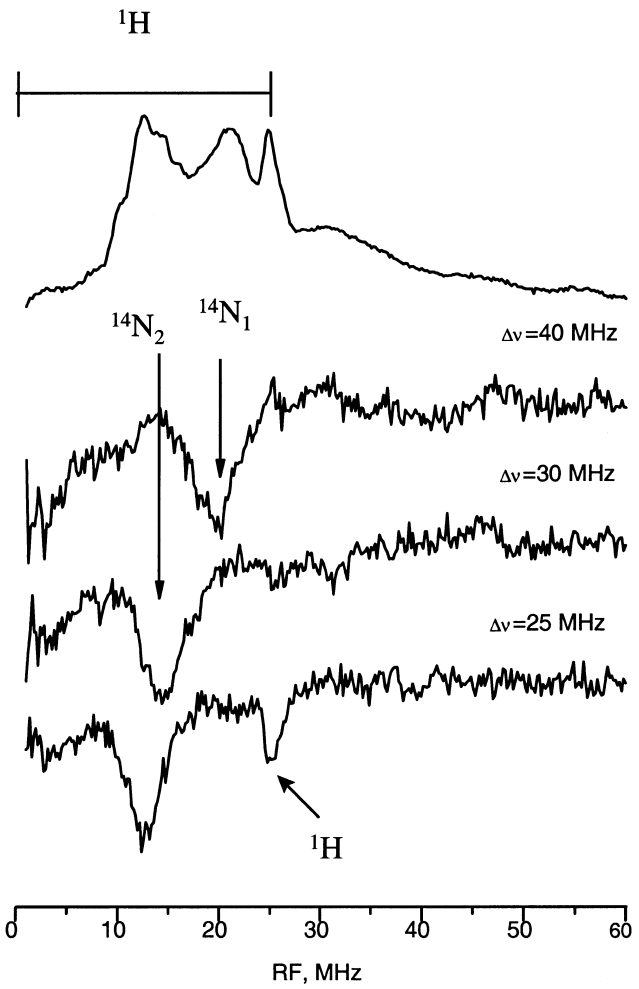


Fig. 9. Davies-ENDOR spectrum of a laccase : azide (1 : 7) solution (1.5 mM, pH = 6) recorded at 4 K. The experimental conditions: ν = 9.40 GHz, B_0 = 3220 G, RF pulse length, 3.5 μs and MW pulse lengths 0.04, 0.02, 0.04 μs. The three bottom traces correspond to hyperfine-selective spectra obtained with microwave pulses of 0.16, 0.08 and 0.16 μs and the Δ-values depicted on the figure.

could be distinguished using the hyperfine-selective (HS) ENDOR pulse sequence [26] as shown in Fig. 9. Selecting $\Delta = \nu_1 - \nu_2 = 40$ MHz generated a signal at 20 MHz, showing that a ^{14}N nucleus with $A_N = 40$ MHz is present, in agreement with the measurements carried out at 3320 G. Another ^{14}N nucleus, with ENDOR frequency of 15 MHz and $A_N \approx 30$ MHz is detected by setting $\Delta = 30$ MHz. It is assigned to histidine ^{14}N nucleus (or nuclei) directly bound to the T1 Cu(II). These nitrogen couplings are in agreement with values of 40 and 33 MHz reported earlier for T1 in laccase [31,32]. The spectrum recorded with $\Delta = 25$ MHz exhibits two signals: one at 12.5 MHz, which is assigned to ^{14}N with a coupling of 25 MHz that may be part of the broad line observed with $\Delta = 30$ MHz. The other line, appearing at 25 MHz, is assigned to the high field component of a proton doublet with $A_H \approx 25$ MHz. This corresponds to the weakly coupled case ($A_H < 2\nu_I$) where the ENDOR signals consist of a doublet with a splitting of A_H centered at ν_I [30]. A 25-MHz coupling is typical for the β-protons of the cysteine ligand in the T1 center [31]. The low frequency component of this doublet is not observed due difficulties in detecting ENDOR signals with very low frequencies.

Davies ENDOR spectra of laccase and laccase : azide (1 : 1) recorded at several field positions along their EPR power patterns were essentially identical. In contrast, small but significant and consistent differences were observed in the spectra of laccase : azide (1 : 7) presented in Fig. 8. Unfortunately, due to the fast relaxation of the $g \approx 1.9$ signal, its relative intensity in the ED-EPR spectrum was too weak to allow ENDOR experiments at the corresponding field even at 2 K. The spectra shown in Fig. 6 were normalized to the intensity of the high frequency ^{1}H signal at ≈ 25 MHz. The latter was used as an internal standard because the T1 Cu(II) is not affected by the azide binding. This normalization, however, could not be carried out for the spectrum recorded at 3320 G where there was no contribution from the T1 site. The right column in Fig. 8 shows the difference spectra, laccase – laccase : azide (1 : 7). Although the spectra of the azide containing sample did not exhibit any new signals, they all show a change in the relative intensities, namely, a decrease in the T2 signal intensity around 20 MHz and a modest increase at 12–15 MHz. The field dependence of the latter is similar to that of the proton's Larmor frequency, thus suggesting that it arises from weakly coupled protons. Although the spectra were recorded with nonselective pulses to reduce the contribution of such protons, they could not be completely eliminated.

We have also carried out electron-spin echo envelope modulation (ESEEM) experiments on the laccase : azide (1 : 7) derivative in an attempt to detect the azide binding to T2. No differences were observed between native laccase and the azide derivative (spectra not shown).

DISCUSSION

The most prominent effects of azide binding on the EPR spectra of laccase and AO are the appearance of a $g < 2$ signal assigned to an $S = 1$, dipolar coupled Cu(II) pair, and the reduction in the T2 signal intensity. The $S = 1$ species is characterized by a relatively fast spin-lattice relaxation rate and therefore it can be observed only below 40 K. The microwave power saturation experiments showed that while the $S = 1$ signal does not reach saturation with maximum power even at 10 K, that of the T1/T2 signal begins to saturate between −6

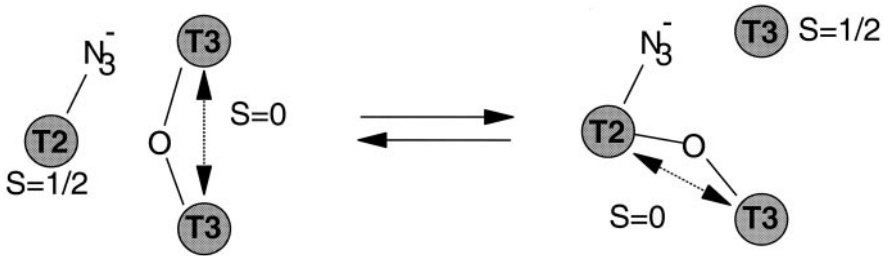


Fig. 10. Possible equilibrium in the magnetic coupling between the three Cu(II) ions in the trinuclear center of the laccase : azide complex.

and −9 dB. As the EPR signal reaches a maximum value at a power level P_{max} for which $S = 1/2$, these different power levels can be used to estimate T_1 of the $S = 1$ Cu(II) pair relative to the T1,T2 Cu(II) ions according to [29]:

$$\frac{P_{max}(T1,T2)}{P_{max}(S=1)} = \frac{H^2_{1,max}(T1,T2)}{H^2_{1,max}(S=1)} = \frac{T_1^2(S=1)}{T_1^2(T1,T2)} \quad (1)$$

In the derivation of eqn (1), an upper limit for T_2 of $T_2 = T_1$ was assumed, yielding $[(T_1(T1,T2))/(T_1(S = 1))] > 3$ at 10 K. Reported T_1 values of T1 and T2 Cu(II) ions in various proteins at 16 K are in the range 30–47 μs and 320–560 μs, respectively [34]. We attribute the faster relaxation of the $g < 2$ signal to its triplet state and its proximity to the paramagnetic T2 Cu(II) ion.

Unfortunately, due to overlap with the signals of T1 and T2, the full line-shape of the $S = 1$ pair could not be resolved. A signal with a similar low g-value, and temperature and power dependencies has been observed in reaction intermediates produced during reoxidation of reduced fungal and tree laccase with O_2 [35,36]. The decoupling of the anti-ferromagnetic interaction within the T3 site by azide has already been observed for laccase [8,11], but not for AO [12] although the 3D structure of the AO azide derivative shows that the bridging oxygen has been removed. The separation of the trinuclear spin system into two $S = 1$ and $S = 1/2$ systems takes place only when two out of the three Cu(II) ions are significantly closer, as indeed found in the 3D structure of the AO : azide derivative (0.51, 0.46, 0.36 nm) [5].

The powder pattern of Cu(II) pairs with $S = 1$ consists of two doublets: an intense one with a splitting of $d_\perp = [D/(g_\perp\beta)]$ and a weaker one with a splitting of $d_\parallel = [2D/(g_\parallel\beta)]$. A third, weaker signal corresponding the double quantum transition, $\Delta M_S = 2$ also appears [36]. The zero field splitting (ZFS) interaction parameter, D, includes a pseudo-dipolar term, D_{ex}, and a dipolar term, D_{dip} according to [38]:

$$D = D_{ex} + D_{dip} \quad (2)$$

where

$$D_{ex} = -\frac{1}{8}J\left[\frac{1}{4}(g_\parallel - 2)^2 - (g_\perp - 2)^2\right] \quad (3)$$

$$D_{dip} = -\left(g_\parallel^2 + \frac{1}{2}g_\perp^2\right)\frac{\beta^2}{r^3} \quad (4)$$

J is the isotropic exchange interaction constant, r is the distance between the two Cu(II) ions and β is the Bohr-magneton. When resolution is high, the doublet exhibits a fine structure due to the ^{63,65}Cu hyperfine interaction as observed in Cu(II) pairs in copper-thorium and copper-cerium oxide catalysts [37,39]. In the latter all spectral features expected from the ZFS interaction, g-anisotropy and hyperfine couplings are easily identified. Comparison of our EPR data with experimental and calculated spectra presented in [39,40] suggests that the $g < 2$ line corresponds to the high field component of the $d_\perp$ doublet. Accordingly, the shift of this signal from the $g_\perp$ position is $^1/_2 d_\perp$ and it can be used to estimate D. This shift is ≈ 330 G in laccase and ≈ 200 G in AO, indicating that the distance between the Cu(II) ions of the pair is shorter in laccase. Using Eqn 4 and assuming that the anisotropic exchange contribution to D is negligible [40], distances of 0.36 nm and 0.42 nm for laccase and AO, respectively, are obtained. The closest Cu–Cu distance in the trinuclear center in the AO-azide derivative is 0.36 nm, between Cu3 and Cu4 [5]. The above calculated longer distance we obtained could be a consequence of neglecting the anisotropic exchange contribution and of possible errors in $g_\perp$ and $g_\parallel$, which were taken to be similar to the values of T2 in the corresponding native proteins.

The EPR spectra of AO : azide (1 : 7) recorded at 130 and 10 K show the presence of a perturbed T2(AO) : N_3^- which has already been reported [12]. Hence, at least part of the decrease in the T2 intensity is attributed to the formation of the perturbed T2(AO) : N_3^-. The relaxation characteristics of T2(AO) : N_3^- are similar to those of the unperturbed T2 in AO showing that it cannot be in the close vicinity of the $S = 1$ Cu(II) pair. Such a proximity would have increased significantly its relaxation rate [41,42]. Accordingly, we conclude that the presence of a sevenfold excess of azide generates two new types of trinuclear centers in AO. In the first azide binds to one of the T3 Cu(II) ions, disrupts the anti-ferromagnetic coupling generating a $S = 1$ Cu(II) pair that enhances the relaxation of the proximal T2 relative to its value in native AO. This faster relaxation rate contributes to the observed reduction of the T2 signal. The formation of this species is consistent with the 3D structure [5]. In the second type, azide binding generates a perturbed T2 [T2(AO) : N_3^-] where the anti-ferromagnetic coupling is not affected. Our results, however, do not allow resolving whether the T3 comprises the same Cu(II) ions as in T3 in native AO. In addition to the two types described, the presence of an unperturbed T2 signal at both 80 and 14 K indicates that either only a fraction of the molecules has reacted with azide or that the azide can also bind to T3 without perturbing the T3 coupling as suggested by one of the models of Cole *et al.* [12].

The temperature and microwave power dependent EPR measurements as well as the ENDOR spectra of the 1 : 7 laccase : azide sample generally reflect the same behavior as found for AO, with the exception that T2(lac) : N_3^- was observed only in the 130 K EPR spectra and not in the 10 K spectra nor in the 4.2 K pulsed ENDOR spectra. As in AO, three species could be distinguished: one that does not bind azide, or where azide binds to T3 without perturbing T2, and is responsible for the observed pulsed ENDOR spectrum. In the second the azide binds to T3 and disrupts its anti-ferromagnetic coupling, forming the $S = 1$ Cu(II) pair and a faster relaxing T2. This species is not observed in the ENDOR spectrum due to its fast relaxation and is responsible for the reduction in the T2 signals in the ENDOR and EPR spectra. In the third species, azide binds to T2, generating T2(lac) : N_3^- without disturbing the T3 anti-ferromagnetic coupling. The disappearance of the spectral features of T2(lac) : N_3^- below 90 K and the absence of any indication in the ENDOR spectra for an azide signal is however, surprising. The features of T2(lac) : N_3^- are also absent from the 8 K spectrum of laccase in the presence of 2.5-fold azide reported earlier [10].

One possible rationale for the absence of the T2(lac) : N_3^- signal at low temperatures is a conformational change that takes place between 130 and 90 K and results in a change in the magnetic coupling among the three copper ions, as shown in Fig. 10. A similar situation was reported for native laccase where two different Cu(II) ions comprise the T3 site at room temperature and at 150 K [33]. Such a change will turn the EPR active T2(lac) : N_3^- at 130 K into a part of the EPR inactive T3 pair below 90 K. One of the T3 Cu(II) ions will then become EPR active and contribute to the low temperature EPR spectrum as well as to the ENDOR spectrum. These spectra indicate that this Cu(II) ion has g-values and ^{63,65}Cu hyperfine coupling similar to those of T2 in native laccase, although it may experience some subtle variation in its close vicinity as evidenced by the small change in the distant proton signal

(at ≈ 13 MHz) in the ENDOR spectrum. This explanation is further supported by a recent study that showed that in the difluoro complex of native laccase the readily replaceable Cu(II) ion in the trinuclear center is not EPR active at low temperatures [43].

Finally, we discuss our results in relation to the models of azide binding to laccase based on earlier MCD and EPR results of Cole *et al*. [11]. In their model the first azide binds to ≈ 90% of the trinuclear centers as a bridge between T2 and one of the T3, yet without disrupting the anti-ferromagnetic coupling between the T3 pair. We found no evidence for the perturbation of the EPR detected Cu(II) by the azide binding below 90 K. In the remaining ≈ 10% of the centers, the mode of azide binding is similar but the anti-ferromagnetic coupling is disrupted. Our results are consistent with this model and the absence of the T2(lac) : N_3^- in the ENDOR spectrum can be rationalized by its faster relaxation rate. The second azide is suggested by Cole *et al*. [11] to bind with a lower affinity as bridging between the T2 and the second T3 Cu(II), again without disrupting the T3 coupling. Our results do not confirm the presence of this species. This, however, can be attributed to the lower concentration of azide we used.

The interaction of the trinuclear cluster in T1Hg *Rhus* laccase with azide at pH 4.5 (as opposed to our study and that of Cole *et al*. [12] where the pH was 6.0) was investigated by EPR at 77 K. Direct azide coordination was deduced from the significant reduction in $A_{\|}$ and EPR simulations of the spectrum of a sample where ^{15}N enriched azide was used [13]. The fine structure of the $A_{\|}$ features could be reproduced with three ^{14}N and one ^{15}N nuclei, all with a hyperfine coupling of 11×10^{-4} cm^{-1} (32 MHz). The observation of the N_3^- coupling is attributed to the structural transition which the trinuclear center undergoes at pH ≈ 4 [13].

In summary, the systematic power and temperature dependence investigation of the EPR spectra of the laccase : azide adducts in combination with the ENDOR measurements allowed us to distinguish between three different types of T2 centers based on their relaxation properties. In addition, the decrease in the T2 signal upon azide binding was found to be, at least partially, due to enhanced relaxation caused by its proximity to the $S = 1$ copper pair. Finally, the disappearance of the perturbed T2 Cu(II) signal at low temperatures suggests a change in the magnetic coupling between the copper ions in the trinuclear center at low temperatures. Nonetheless, many questions remain open and although 3D structures are available for several AO derivatives [3–5,14] and for one laccase derivative [17], the magnetic and structural properties of the trinuclear : azide adduct are still not sufficiently understood and the relation between the various spectroscopic observations and the 3D structure of the azide derivative is not yet fully established. The overlapping signals and the fast relaxation of the $S = 1$ Cu(II) pair prevents detailed pulsed EPR/ENDOR studies that would otherwise directly demonstrate hyperfine interaction between the Cu(II) and the bound azide.

CONCLUSIONS

The interaction of the trinuclear copper centers in laccase and ascorbate oxidase with azide generates in each three different species: (a) In the first, the azide binds to one or both of the Cu(II) ions of the T3 pair and disrupts its anti-ferromagnetic coupling. This yields a dipolar coupled Cu(II) pair with $S = 1$ that enhances the spin lattice relaxation rate of the proximal T2 Cu(II). The distance between the Cu(II) ions comprising the $S = 1$ pair is shorter in laccase than in AO. (b) In the second type of species the azide binds to T2 Cu(II) without disturbing the T3 coupling. A perturbed T2 site is evident in the AO : azide sample in the range 14–80 K whereas in laccase : azide it is not observed below 90 K. This suggests that the magnetic coupling among the Cu(II) ions within the trinuclear center in the laccase-azide derivative is temperature dependent. (c) The third species represents centers where azide does not bind at all or where it binds to T3 without causing changes in the EPR and ENDOR spectra. Recent X-ray structure determination of the ceruloplasm–azide complex has shown that the azide ion binds to one of the T3 copper ions and the oxygen bridging the T3 copper ions is lost [44].

ACKNOWLEDGEMENTS

The research reported in the paper has been supported by the German Israeli Science Foundation.

REFERENCES

1. Malkin, R. & Malmström B.G. (1990) The state and function of copper in biological systems. *Adv. Enzymol.* **33**, 177–244.
2. Reinhammar, B. (1982) Laccase. In *Copper Proteins and Copper Enzymes* (Lontie, R., ed.), Vol. 3, pp. 1–35. CRC Press, Boca Raton, FL, USA.
3. Messerschmidt, A., Rossi, A., Ladenstein, R., Huber, R., Bolognesi, M., Gatti, G., Marchesini, A., Petruzzelli, R. & Finazzi-Agro, A. (1989) X-ray crystal structure of the blue oxidase ascorbate oxidase from zuchini. Analysis of the polypeptide fold and a model of the copper sites and ligands. *J. Mol. Biol.* **206**, 513–529.
4. Messerschmidt, A., Ladenstein, R., Huber, R., Bolognesi, M., Avigliano, L., Petruzzelli, R., Rossi, A. & Finazzi-Agro, A. (1992) Refined crystal structure of ascorbate oxidase at 1.9 Å resolution. *J. Mol. Biol.* **224**, 179–205.
5. Messerschmidt, A., Luecke, H. & Huber, R. (1993) X-ray structures and mechanistic implications of three functional derivatives of ascorbate oxidase from zucchini. *J. Mol. Biol.* **230**, 997–1014.
6. Farver, O., Goldberg, M. & Pecht, I. (1978) Circular dichroic spectrum of the laccase-peroxidase derivative. *FEBS Lett.* **94**, 383–386.
7. Marchesini, A. & Kroneck, P.M.H. (1979) Ascorbate oxidase from *Cucurbita pepamedullosa*. New method of purification and reinvestigation of properties. *Eur. J. Biochem.* **101**, 65–76.
8. Morpurgo, L., Desideri, A. & Rotilio, G. (1982) Heterogeneity of the type 3 copper in Japanese-lacquer-tree (*Rhus vernicifera*) laccase. *Biochem. J.* **207**, 625–627.
9. Allendorf, M.D., Spira, D.J. & Solomon, E.I. (1985) Low-temperature magnetic circular dichroism studies of native laccase. spectroscopic evidence for exogenous ligand bridging at a trinuclear copper active site. *Proc. Natl Acad. Sci. USA* **82**, 3063–3067.
10. Spira-Solomon, D.J., Allendorf, M.D. & Solomon, E.I. (1986) Low-temperature magnetic circular dichroism studies of native laccase. Confirmation of a trinuclear copper active site. *J. Am. Chem. Soc.* **108**, 5318–5328.
11. Cole, J.L., Clark, P.A. & Solomon, E.I. (1990) Spectroscopic and chemical studies of the laccase trinucelar copper active site. Geometric and electronic structure. *J. Am. Chem. Soc.* **112**, 9534–9548.
12. Cole, J.L., Avigliano, L., Morpurgo, L. & Solomon, E.I. (1991) Spectroscopic and chemical studies of the ascorbate oxidase trinuclear copper active site. Comparison to laccase. *J. Am. Chem. Soc.* **113**, 9080–9089.
13. Peyratout, C.S., Holm, S.R. & McMillin, D.R. (1994) EPR studies of ligand binding to the type 2/type 3 cluster in tree laccase. *Arch. Biochem. Biophys.* **314**, 405–411.
14. Messerschmidt, A., Steigmann, W., Huber, R., Lang, G. & Kroneck, P.M.H. (1992) X-ray crystallographic characterization of type-2-depleted ascorbate oxidase from zucchini. *Eur. J. Biochem.* **209**, 597–602.

15. Frank, P., Farver, O. & Pecht, I. (1983) The type 3 copper site is intact but labile in type 2 depleted laccase. *J. Biol. Chem.* **258**, 11112–11117.
16. Messerschmidt, A. & Huber, R. (1990) The blue oxidases, ascorbate oxidase, laccase and ceruloplasmin. *Eur. J. Biochem.* **187**, 314–352.
17. Ducros, V., Brzozowski, A.M., Wilson, K.S., Brown, S.H., Østergaard, P., Schneider, P., Yaver, D.S., Pedersen, A.H. & Davies, G.J. (1998) Crystal structure of the type-2 Cu depleted laccase from Coprinus cinereus at 2.2 Å resolution. *Nat. Struct. Biol.* **5**, 310–316.
18. Bränden, R., Deinum, J. & Coleman, M. (1978) A mass spectrometric investigation of the reaction between $^{18}O_2$ and reduced tree laccase. A differentiation between two water molecules formed. *FEBS Lett.* **89**, 180–182.
19. Bränden, R. & Deinum, J. (1978) The effect of pH on the oxygen intermediate and the dioxygen reducing site in blue oxidases. *Biochem. Biophys. Acta* **524**, 297–304.
20. Farver, O., Goldberg, M. & Pecht, I. (1980) A circular dichroism study of the reactions of Rhus laccase with dioxygen. *Eur. J. Biochem.* **104**, 71–77.
21. Hirota, S., Matsumoto, H., Huang, H.-W., Sakurai, T., Kitagawa, T. & Yamauchi, O. (1998) Observation of Cu-N_3^- stretching and N_3^- asymmetric stretching bands for mono-Azide adduct of *Rhus vernicifera* laccase. *Bioch. Biophys. Res. Communications* **243**, 435–437.
22. Reinhammar, B. (1970) Purification and properties of laccase and stellacyanin from *Rhus vernicifera*. *Bioch. Bioph. Acta* **205**, 35–47.
23. Goldfarb, D., Fauth, J.-M., Tor, Y. & Shanzer, A. (1991) Study of Cu(II) binding to chiral tripodal ligands by electron spin echo spectroscopy. *J. Am. Chem. Soc.* **113**, 1941–1948.
24. Shane, J.J., Gromov, I., Vega, S. & Goldfarb, D. (1998) A versatile pulsed X-band ENDOR spectrometer. *Rev. Sci. Instrum.* **69**, 3357–3363.
25. Forrer, J., Pfenninger, S., Eisengger, J. & Schweiger, A. (1990) A pulsed ENDOR probehead with the bridged loop-gap resonator. Construction and performance. *Rev. Sci. Instrum.* **61**, 3360–3367.
26. Thomann, H. & Bernardo, M. (1990) Pulsed electron-nuclear-electron triple resonance spectroscopy. *Chem. Phys. Lett.* **169**, 5–11.
27. Davies, E.R. (1974) A new pulse ENDOR technique. *Phys. Lett.* **47A**, 1–2.
28. Atherto, N.M. (1993) *Principle of Electron Spin Resonance*, pp. 283–291. Ellis Horwood PTR, Prentice Hall, NewYork, USA.
29. Poole, C.P. (1996) *Electron Spin Resonance, a Comprehensive Treaties on Experimental Techniques*, 2nd edn, pp. 589–600. Dover Publications, Inc., Mineola, New York, USA.
30. Gemperle, C. & Schweiger, A. (1991) Pulsed electron-nuclear double resonance methodology. *Chem Rev.* **91**, 1481–1505.
31. Werst, M.M., Davoust, C.E. & Hoffman, B.M. (1991) Ligand spin densities in blue copper proteins by Q-band ^1H and ^{14}N ENDOR spectroscopy. *J. Am. Chem. Soc.* **113**, 1533–1538.
32. Roberts, J.E., Cline, J.F., Lum, V., Freeman, H., Gray, H.B., Peisach, J., Reinhammar, B. & Hoffman, B.M. (1984) Comparative ENDOR study of six blue copper proteins. *J. Am. Chem. Soc.* **106**, 5324–5330.
33. Li, J.-B., McMillin, D.R. & Antholine, W. (1992) Evidence for temperature-dependent changes in the coupling within the type 2/type 3 cluster of laccase. *J. Am. Chem. Soc.* **114**, 725–727.
34. Pfenninger, S., Antholine, W.E., Barr, M.E., Hyde, J.S., Kroneck, P.M.H. & Zumft, W.G. (1995) Electron spin lattice relaxation of the [Cu(1.5)…Cu(1.5)] dinuclear copper center in nitrous oxide reductase. *Biophys. J.* **69**, 2761–2769.
35. Aasa, R., Bränden, R., Deinum, B.J., Malmström, B.G., Reinhammar, B. & Vänngård, T. (1976) A paramagnetic intermediate in the reduction of oxygen by reduced laccase. *FEBS Lett.* **61**, 116–119.
36. Frank, P. & Pecht, I. (1986) Redox titrations of type 2 copper depleted *Rhus* laccase. reductive decoupling and oxidative recognition of the type 3 site. *J. Phys. Chem.* **90**, 3809–3814.
37. Bechara, R., d'Husser, A., Aissi, C.F., Guelton, M., Bonnelle, J.P. & Aboukais, A. (1990) Preparation and characterization of copper-thorium oxide catalysts. 2. ESR study of copper (II) ion pairs and correlation with the selective hydrogenation of isoprene. *Chem. Mater.* **2**, 522–526.
38. Smith, T.D. & Pilbrow, J.R. (1974) The determination of structural properties of dimeric transition metal ion complexes from EPR spectra. *Coord. Chem. Rev.* **13**, 173–278.
39. Aboukais, A., Bennani, A., Aissi, C.F., Wrobel, G. & Vedrin, J.C. (1992) Highly resolved electron paramagnetic resonance spectrum of copper (II) ion pairs in CuCe oxide. *J. Chem. Soc. Farad. Trans.* **88**, 615–620.
40. Wilcox, D.E., Long, J.R. & Solomon, E.I. (1984) EPR studies of the 'EPR nondetectable' Met derivative hemocyanin. Perturbations and displacement of the endogenous bridge in the coupled binuclear copper active site. *J. Am. Chem. Soc.* **106**, 2186–2194.
41. Kulikov, A.V. & Likhtenshtein, G.I. (1977) The use of spin relaxation phenomena in the investigation of the structure of model and biological systems by the method of spin labels. *Adv. Mol. Relax. Interac. Processes* **10**, 47–69.
42. Budker, V., Du, J.-L., Seiter, M., Eaton, G.R., Eaton, S. & S. Electron-electron spin–spin interaction in spin-labeled low-spin methemoglobin. *Biophys. J.* **68**, 2531–2542.
43. Fraterrigo, T.L., Miller, C., Reinhammar, B. & McMillin, D.R. (1999) Which copper is paramagnetic in the type 2/type 3 cluster of laccase? *J. Biol. Inorg. Chem.* **4**, 183–197.
44. Zaitsev, V.N., Zaitseva, I., Papiz, M. & Lindley, P.F. (1999) An X-ray crystallographic study of the binding sites of the azide inhibitor and organic substrates to ceruloplasmin, a multi-copper oxidase in the plasma. *J. Biol. Inorg. Chem.* **4**, 579–587.